

Sovereign Heavy Industries — Quality Audit Report

CONFIDENTIAL INDUSTRIAL INSPECTION • ASSET TRB-702-U4 • DATE: 2026-09-08

1. Executive Audit Summary

This official inspection report covers Turbomachinery Unit 4 (Asset ID: TRB-702-U4).
Routine quarterly maintenance testing was executed in Sector 7 under steady-state operating conditions.
Sensor telemetry logs were recorded across all critical bearings, rotor shafts, and lube subsystems.

2. Preliminary Observations & Scope

During inspection, telemetry instruments flagged severe thermal excursion on Radial Bearing #3.
Detailed field worksheet and sensor telemetry recordings are attached on Page 2 (Scanned Worksheet).
Governing Standard: SOP-IND-702 Turbomachinery Operating Thresholds Standard.

Asset ID: TRB-702-U4 | Unit 4 Turbomachinery Inspection Report

Inspection Date: 2026-09-08 | Shift: Morning Quality Audit (08:00 - 12:00)

Lead Inspector: Dr. R. K. Patel (Chief Reliability Inspector, Level III)

COMPONENT	BASELINE	MEASURED TEMP	VIBRATION	STATUS
Radial Bearing #1	70.0 C	71.2 C	2.1 mm/s	PASS (NORMAL)
Radial Bearing #2	70.0 C	73.5 C	2.4 mm/s	PASS (NORMAL)
Radial Bearing #3	70.0 C	88.4 C	4.8 mm/s	CRITICAL EXCEEDANCE
Thrust Bearing #4	68.0 C	69.8 C	1.9 mm/s	PASS (NORMAL)
High-Pressure Stator	180.0 C	182.4 C	1.2 mm/s	PASS (NORMAL)
Lube Oil Sump Tank	52.0 C	54.1 C	N/A	PASS (NORMAL)

FIELD ENGINEER NOTES & ANOMALY ASSESSMENT:

- Radial Bearing #3 recorded temperature of 88.4 C under sustained 100% continuous load.
- Standard SOP-IND-702 limit is 75.0 C. Thermal delta exceedance is +13.4 C.
- Radial vibration on Bearing #3 reached 4.8 mm/s, breaching ISO-10816 threshold (3.5 mm/s).
- Immediate Level 1 controlled shutdown and unit isolation required within 12 hours.
- Formal Approval Note must be signed by Chief Plant Engineer prior to restart.

AUDIT VERIFIED

LEVEL 1 CRITICAL

DATE: 2026-09-08

SOVEREIGN QA/QC